

Nonlinear State Estimation of a Pendulum Using Vision Measurements

A comparative study of EKF, UKF, and Particle Filtering

Description

This project addresses the problem of estimating the full state of a pendulum system using nonlinear dynamics and camera-based measurements. The system consists of a ball attached to a string, observed by a monocular camera that provides only image-space measurements.

The estimation problem is challenging due to:

- nonlinear dynamics with multiple coupled forces
- nonlinear measurement model arising from perspective projection
- partial observability of the system (unknown hook position)

To study this problem, three Bayesian estimation methods are implemented and compared:

- Extended Kalman Filter (EKF)
- Sigma-point Kalman Filter (UKF)
- Particle Filter (PF)

This project explores how different estimation strategies perform on the same nonlinear system, highlighting trade-offs between accuracy, robustness, and computational complexity.

System Model

The state vector and its dynamics are defined as

$$\underline{x} = [\underline{p} \quad \underline{v} \quad \underline{h}]^\top$$

$$\dot{\underline{x}} = \begin{bmatrix} \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} \underline{x} + \begin{bmatrix} \mathbf{0}_3 \\ \underline{a} \\ \mathbf{0}_3 \end{bmatrix}$$

where $\underline{p}$ is the position of the ball, $\underline{v}$ is its velocity, and $\underline{h}$ is the position of the hook.

The dynamics include gravitational acceleration, string tension effects, and aerodynamic drag:

$$\underline{a} = \underline{a}_g + \underline{a}_c + \underline{a}_d + \underline{g}$$

Where we discuss each acceleration component below.

Counter-gravitational Acceleration

We are assuming a non-elastic string, so any force gravity puts on the ball will be countered by the string. The position of the ball, $\underline{p}$, relative to the hook, $\underline{h}$, is

$$\underline{s} = \underline{p} - \underline{h}$$

Its unit vector is $\underline{u}_s = \underline{s}/s$, where $s = \|\underline{s}\|$. So, the tension acceleration (counter-gravitational) on the string due to gravity is

$$\underline{a}_g = -\underline{u}_s(\underline{u}_s \cdot \underline{g})$$

Centripetal Acceleration

Another tension on the string is due to the velocity, $\underline{v}$, of the ball. This is the centripetal acceleration.

$$\underline{a}_c = -\underline{u}_s \frac{v^2}{s}$$

where $v = \|\underline{v}\|$.

Drag Acceleration

As the ball moves through the air, there is a drag which opposes its motion. This acceleration is defined in the opposite direction to the velocity and its magnitude is the velocity squared scaled by the drag coefficient, k_d

$$\underline{a}_d = -\underline{v}(k_d v)$$

Discrete Space Dynamics

In discrete space, the dynamics equations are

$$\underline{x}_{k+1} = \underline{\phi}(\underline{x}_k)$$

$$\underline{\phi}(\underline{x}) = \mathbf{I}_9 \underline{x} + T \begin{bmatrix} \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} \underline{x} + \begin{bmatrix} \frac{1}{2} T^2 \underline{a} \\ T \underline{a} \\ \underline{0}_3 \end{bmatrix}$$

Dynamics Jacobians

The Extended Kalman Filter requires the use of Jacobians to approximate how uncertainty in the state propagates through nonlinear dynamics by linearizing these dynamics. Below we will derive these Jacobians.

The Jacobian of the counter-gravitational acceleration is composed of

$$\frac{\partial \underline{a}_g}{\partial p_x} = -2 \frac{s_x}{s^2} \underline{a}_g - \frac{1}{s^2} \begin{bmatrix} 2g_x s_x + g_y s_y + g_z s_z \\ g_x s_y \\ g_x s_z \end{bmatrix}$$

$$\frac{\partial \underline{a}_g}{\partial p_y} = -2 \frac{s_y}{s^2} \underline{a}_g - \frac{1}{s^2} \begin{bmatrix} g_y s_x \\ g_x s_x + 2g_y s_y + g_z s_z \\ g_y s_z \end{bmatrix}$$

$$\frac{\partial \underline{a}_g}{\partial p_z} = -2 \frac{s_z}{s^2} \underline{a}_g - \frac{1}{s^2} \begin{bmatrix} g_z s_x \\ g_z s_y \\ g_x s_x + g_y s_y + 2g_z s_z \end{bmatrix}$$

The Jacobian of the centripetal acceleration is composed of

$$\frac{\partial \underline{a}_c}{\partial p_x} = -\frac{v^2}{s^2} \begin{pmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 2 \frac{s_x}{s^2} \underline{s} \end{pmatrix} \quad \frac{\partial \underline{a}_c}{\partial v_x} = -2 \frac{s}{s^2} v_x$$

$$\frac{\partial \underline{a}_c}{\partial p_y} = -\frac{v^2}{s^2} \begin{pmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 2 \frac{s_y}{s^2} \underline{s} \end{pmatrix} \quad \frac{\partial \underline{a}_c}{\partial v_y} = -2 \frac{s}{s^2} v_y$$

$$\frac{\partial \underline{a}_c}{\partial p_z} = -\frac{v^2}{s^2} \begin{pmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - 2 \frac{s_z}{s^2} \underline{s} \end{pmatrix} \quad \frac{\partial \underline{a}_c}{\partial v_z} = -2 \frac{s}{s^2} v_z$$

The Jacobian of the drag acceleration is composed of

$$\frac{\partial \underline{a}_d}{\partial v} = -\frac{k_d}{v} (v^2 \mathbf{I}_3 + \underline{v} \underline{v}^\top)$$

Notice that these Jacobians have only been with respect to the position and velocity of the ball. The Jacobians with respect to the hook position are simply the negative of the Jacobians with respect to the ball position.

Putting all these together, the Φ matrix is

$$\Phi = \mathbf{I}_9 + T \begin{bmatrix} \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{2} T^2 \frac{\partial \underline{a}}{\partial \underline{x}} \\ T \frac{\partial \underline{a}}{\partial \underline{x}} \\ \mathbf{0}_{3 \times 9} \end{bmatrix}$$

The Φ matrix is the discretized dynamics Jacobian. The $\partial \underline{a} / \partial \underline{x}$ is a 3×9 matrix. The first three columns are with respect to the ball position, the next three with respect to the ball velocity, and the last three with respect to the hook position. Put columns vectors like $\partial \underline{a}_g / \partial p_x$ into the first column, $\partial \underline{a}_c / \partial v_x$ into the fourth column, and so on. Vectors like $\partial \underline{a}_g / \partial p_x$ and $\partial \underline{a}_c / \partial p_x$ would add.

Measurement Model

The camera measurement vector is

$$\underline{z} = [r_p \quad u \quad v]^\top$$

Measurements are obtained from a monocular camera. The sensor provides:

- pixel coordinates (u, v)
- apparent radius r_p

These measurements are related to the 3D position through a nonlinear projection model:

$$r_p = \kappa \frac{r_b}{p_z}$$

$$u = c_x + \kappa \frac{p_x}{p_z}$$

$$v = c_y + \kappa \frac{p_y}{p_z}$$

Where r_b is the true radius of the ball, κ is a scaling factor, c_x is the horizontal center of the camera image in pixels, and c_y is the vertical center of the image in pixels. The measurement model introduces strong nonlinearities due to division by depth (p_z).

Measurement Model Jacobians

Like the dynamics, the measurement model requires the use of Jacobians in predicting how uncertainty in the state affects the predicted measurement. Below we will derive the measurement Jacobian.

The measurement Jacobian is composed of

$$\frac{\partial \underline{h}}{\partial \underline{p}} = \begin{bmatrix} 0 & 0 & -\kappa \frac{r_b}{p_z^2} \\ \frac{\kappa}{p_z} & 0 & -\kappa \frac{p_x}{p_z^2} \\ 0 & \frac{\kappa}{p_z} & -\kappa \frac{p_y}{p_z^2} \end{bmatrix}$$

Estimation Methods

Three Bayesian estimation approaches are applied to the same system.

Extended Kalman Filter (EKF)

The EKF linearizes the dynamics and measurement models using analytical Jacobians. This provides computational efficiency but introduces approximation error in highly nonlinear regions.

Sigma-Point Kalman Filter (UKF)

The sigma-point Kalman filter approximates nonlinear transformations using deterministically selected sigma points. This avoids explicit linearization and more accurately captures uncertainty propagation.

Particle Filter (PF)

The particle filter represents the state distribution using weighted samples. This allows it to capture non-Gaussian and multimodal distributions but requires significantly more computation.

Initialization and Implementation

Initial position estimates are obtained by inverting the measurement model. Velocity is estimated using finite differences between early measurements.

The hook position is initialized based on prior geometric assumptions. Covariance matrices and noise parameters are tuned empirically to achieve stable and accurate estimation.

Each filter is implemented using the same system and measurement models to enable direct comparison.

Results

The performance of each estimator is evaluated using trajectory reconstruction and measurement consistency.

Trajectories

Figures 1–3 show the estimated trajectories for each method.

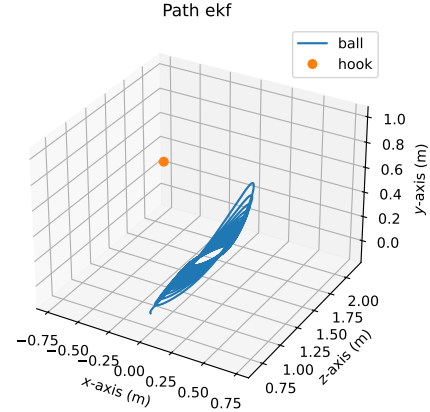


Figure 1: Estimated trajectory for EKF

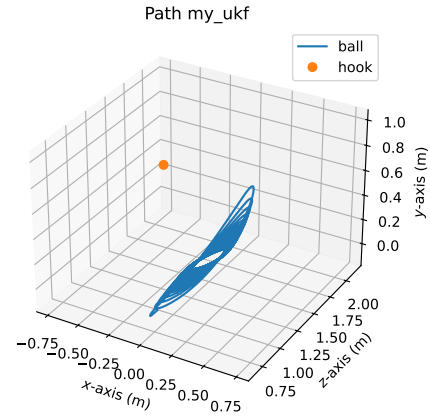


Figure 2: Estimated trajectory for UKF

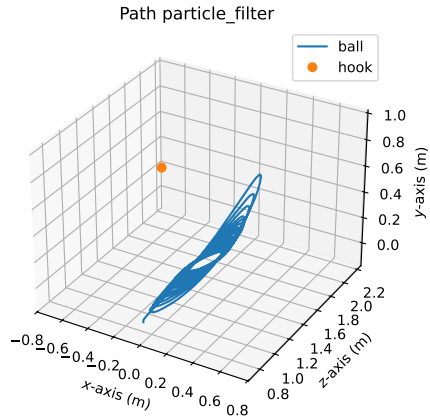


Figure 3: Estimated trajectory for PF

Figures 4–6 show the estimated ball positions for each method.

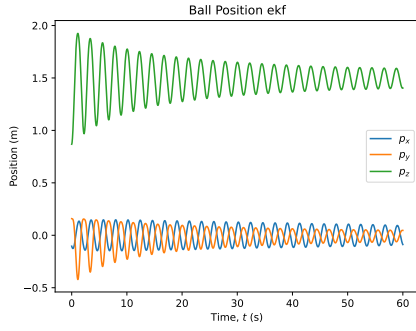


Figure 4: Estimated ball position for EKF

Figures 7–9 show the estimated ball velocities for each method.

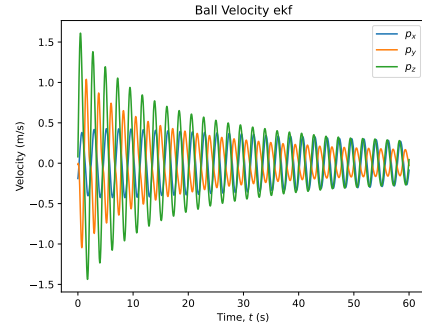


Figure 7: Estimated ball velocity for EKF

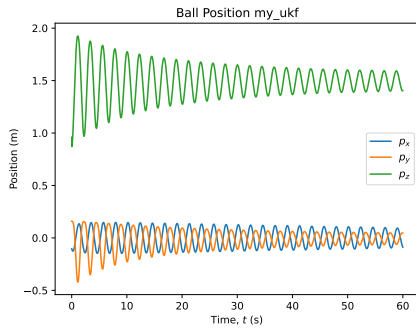


Figure 5: Estimated ball position for UKF

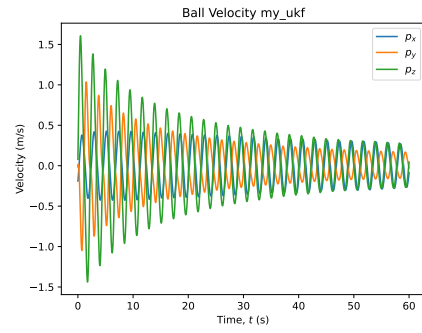


Figure 8: Estimated ball velocity for UKF

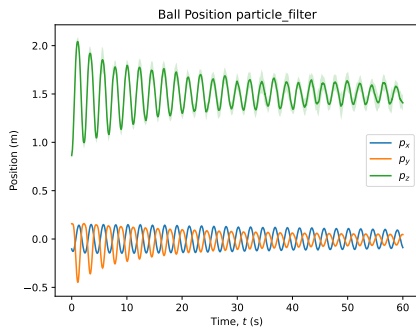


Figure 6: Estimated ball position for PF

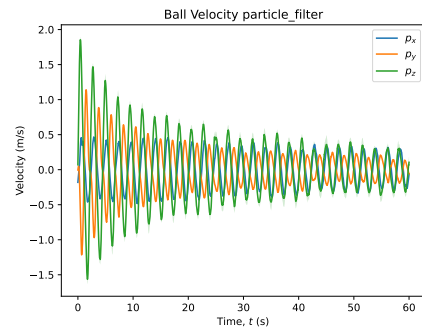


Figure 9: Estimated ball position for PF

Figures 10–12 show the estimated hook positions for each method.

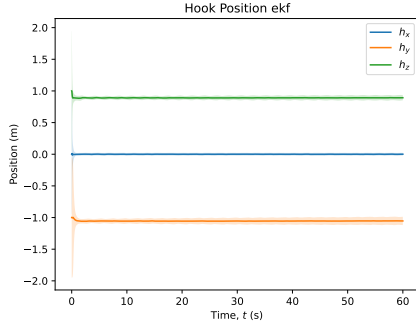


Figure 10: Estimated hook position for EKF

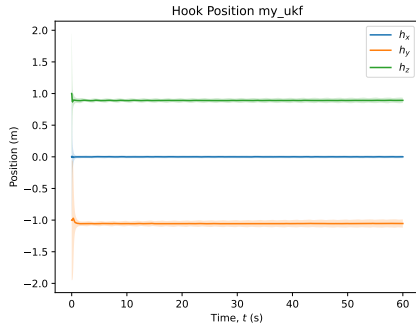


Figure 11: Estimated hook position for UKF

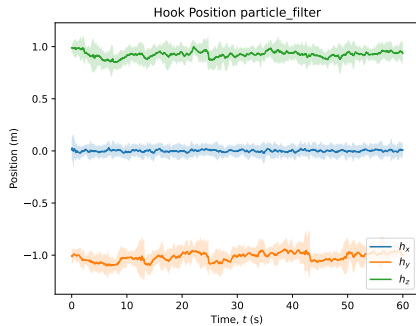


Figure 12: Estimated hook position for PF

IOU

Performance is also evaluated using the IOU metric between predicted and observed image measurements. This function calculates the intersect-over-union (IOU) of the expected pixel circle and the measured pixel circle. If there is a perfect overlap, the IOU value will be 1. Anything else will return less than 1 (minimum 0).

Table 1 shows the average IOU over the duration of the simulation for each filtering method.

Method	IOU
EKF	0.99733
UKF	0.99715
PF	0.97059

Table 1: Intersection-over-union (IOU) values

Discussion

The results highlight clear differences in performance across the three estimation methods.

The EKF achieved the highest IOU, indicating that for this problem the linearization-based approach was sufficiently accurate. This suggests that, despite the nonlinear measurement model, the system operates in a manner where local linear approximations remain valid over each update step.

The UKF performed very similarly to the EKF, with only a slightly lower IOU. This indicates that the additional complexity of sigma-point propagation did not provide a significant advantage in this specific scenario. However, the UKF still offers improved robustness to stronger nonlinearities and does not require analytical Jacobians, which makes it a more flexible approach for more complex systems.

The particle filter produced a lower IOU compared to both Kalman filter approaches, though its performance remained strong overall. Achieving this level of accuracy required approximately 500 particles, highlighting the trade-off between computational cost and estimation quality. While the particle filter is capable of handling highly nonlinear and non-Gaussian systems, its increased computational burden makes it less efficient for problems where Gaussian assumptions are reasonable.

An important observation is that all three filters were able to infer the full 9-dimensional state from

indirect camera measurements. This demonstrates that the combination of system dynamics and measurement structure provides sufficient information to reconstruct the underlying state, even when certain variables (such as depth and hook position) are not directly measured.

Overall, the results suggest that for this problem, the system is sufficiently well-behaved that Gaussian-based filters (EKF and UKF) are both accurate and efficient, while the particle filter provides additional robustness at a higher computational cost.

Conclusion

This project investigated nonlinear state estimation for a pendulum system using vision-based measurements. Three estimation methods were implemented and compared: an Extended Kalman Filter, a Sigma-Point Kalman Filter (UKF), and a Particle Filter.

The results show that all three methods are capable of accurately reconstructing the system state, despite the nonlinear dynamics and indirect measurements. The EKF achieved the highest performance in terms of IOU, with the UKF performing nearly as well. The particle filter also produced accurate estimates but required significantly greater computational effort to achieve comparable performance.

These results highlight an important principle in state estimation: the choice of estimator should be guided by the level of system nonlinearity and the required computational efficiency. For systems where nonlinearities are moderate and Gaussian assumptions are reasonable, Kalman filter-based methods provide an effective and efficient solution. More general methods such as particle filters become advantageous when these assumptions break down.

This project demonstrates the full estimation pipeline for a nonlinear, partially observed system, including modeling, measurement inversion, filter design, and performance evaluation. It also illustrates how different estimation approaches behave when applied to the same problem, providing insight into their relative strengths and limitations.